Agentic Statistical Analysis Assistant

CS7327 | Midterm writeup | Jingcheng Zhou and Junliang Lyu

What we built

We built a local English-language assistant for users who can state a research question but need help choosing and interpreting a statistical analysis. It accepts a CSV and a question, clarifies variable roles and study design, runs a supported test and explains the evidence. Our Python LangGraph StateGraph has 17 nodes and supports Welch independent-samples t-tests, Pearson correlation, chi-square tests of independence and simple ordinary least-squares regression. The current interface is a command line or notebook. Its main unresolved problem is the reliability of interpretation review.

Agent roles and why they are separate

Model A uses Qwen3:8b through Ollama to plan the analysis, assess the meaning of variables and draft the explanation. Model B uses Gemma3:12b to review that explanation against the original question, confirmed design and raw statistical evidence. These are two distinct models. Shared graph state passes the plan, diagnostics, statistics and review feedback between their roles. Model B can request a revision or withhold an explanation that remains unacceptable.

LLM checks of the research goal, variable meaning and design run in parallel with Python checks of numerical suitability; the graph then merges both branches. Python validates the plan and calculates the statistics. A deterministic check verifies numerical placeholders before Model B reviews the interpretation. Language understanding belongs in the models, while calculations and repeatable checks remain in code.

One end to end walkthrough

Our tutoring demonstration uses 80 synthetic student records, with 40 in each group. The question asks whether mean exam score differs between students with and without tutoring. The plan identifies score as the continuous outcome and tutoring as the two-group variable. Supplied teaching metadata describe independent observations and nonrandom group assignment; those facts cannot be inferred from the values alone.

The plan specifies Welch for a difference between two independent group means without requiring equal variances. Python checks usable values, group sizes, variation and distribution warnings. The hypotheses are H0: mean difference = 0 and H1: mean difference ≠ 0, with $\alpha = 0.05$. These checks cannot prove independence, and synthetic data provide a demonstration rather than evidence about actual students.

SciPy gives mean scores of 75.9525 with tutoring and 69.0350 without tutoring. The difference, tutoring minus no tutoring, is 6.9175 points, with 95% CI [2.9689, 10.8661], $t = 3.4927$, $df = 71.4775$ and $p = 0.000824$. We reject H0 under the stated assumptions. The report describes an observed association; nonrandom assignment prevents a causal claim. The graph permits one plan revision, one report revision, two answered clarification rounds and at most 15 model calls.

Follow up questions and memory

A request to explain the result again reuses stored statistics and warnings. A request for a 99% confidence interval recomputes the interval as [1.6763, 12.1587] for this example, while retaining the hypotheses and $\alpha = 0.05$. A new research question returns to planning. Date subsets currently require a newly supplied CSV; automatic date filtering is not implemented. Memory lasts only for the running graph process. Changed data require fresh diagnostics.

What basic statistical checks mean here

The workflow checks relevant column types, missing or nonfinite values, usable sample sizes and variable variation, with method-specific diagnostics and warnings. Chi-square checks expected cell frequencies; unsupported sparse tables can stop analysis. When problem data require correction or exclusion, the assistant pauses for explicit user confirmation and a reason. It preserves the original data and records the modification. Paired or clustered designs, multivariable regression and automatic searches for alternative tests remain outside the supported scope. A warning or passing diagnostic does not establish that the statistical assumptions hold.

A genuine breaking scenario

In earlier live local runs, Pearson and regression produced correct statistics but their reports were withheld after the allowed revision. A literal zero in the null hypothesis conflicted with the numerical placeholder policy. This was an observed integration failure: the arithmetic succeeded, but the assistant failed to deliver an acceptable explanation.

We repaired the handling of null hypotheses, decisions and effect direction, including the distinction between a null slope of zero and an estimated slope. The latest targeted reruns delivered Pearson after one revision and regression without revision. Reviewer false alarms about rounding have also occurred. These repairs improve delivery, but disagreement between two models still does not establish which interpretation is correct.

Evaluation and what the evidence supports

All 93 automated mock tests pass, and the current demo and submission notebooks completed their 14 and 15 code cells in mock mode. Historical real local two-model runs delivered four supported methods and two follow-ups across initial runs and targeted reruns. This is evidence of execution, not a single fresh full run or a model accuracy rate.

Independent human scoring has not been completed. Next, we will evaluate held-out ordinary, ambiguous and failing cases using an independent rubric for method choice, target, numerical fidelity, uncertainty, missing-data handling and causal restraint. Reviewers will record misleading sentences and reasons before resolving disagreements. We will also log reviewer false positives and missed errors. Our strongest evidence is reproducible numerical execution; interpretation reliability remains the main open question.